

HTTPS and The TLS Handshake: Securing the Web

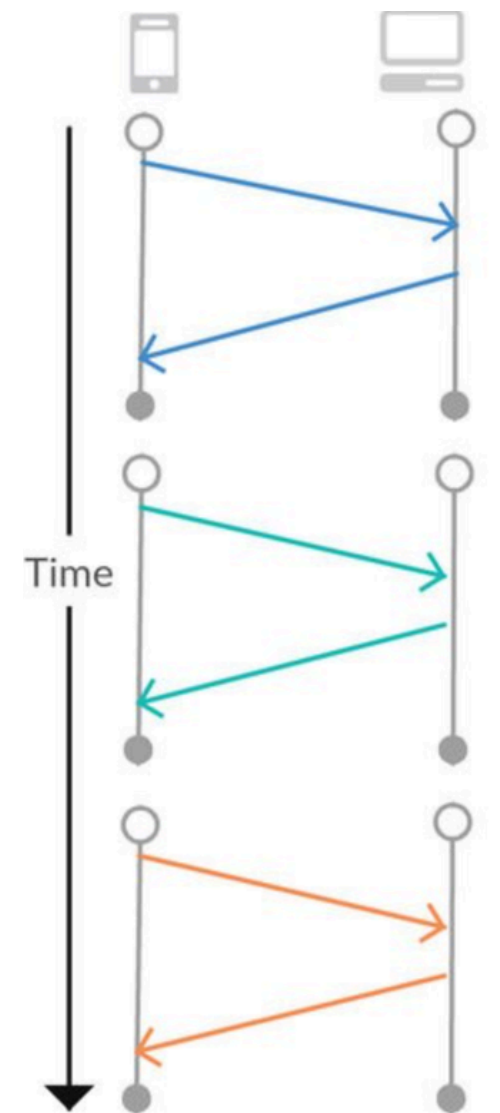
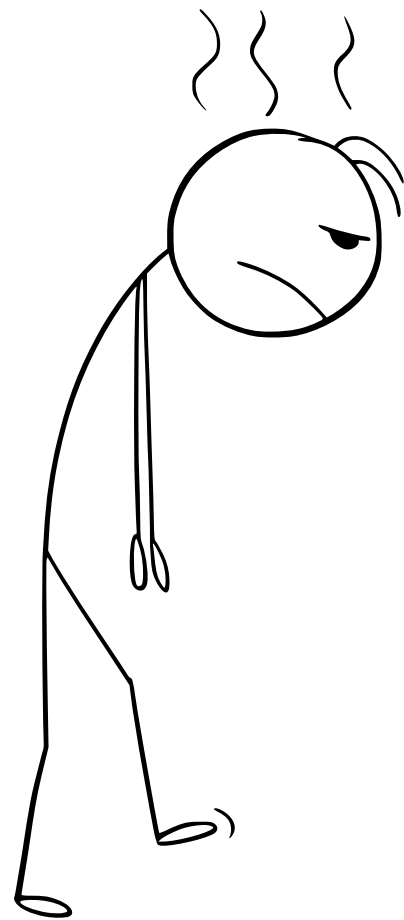
CSAI321:Computer Networks

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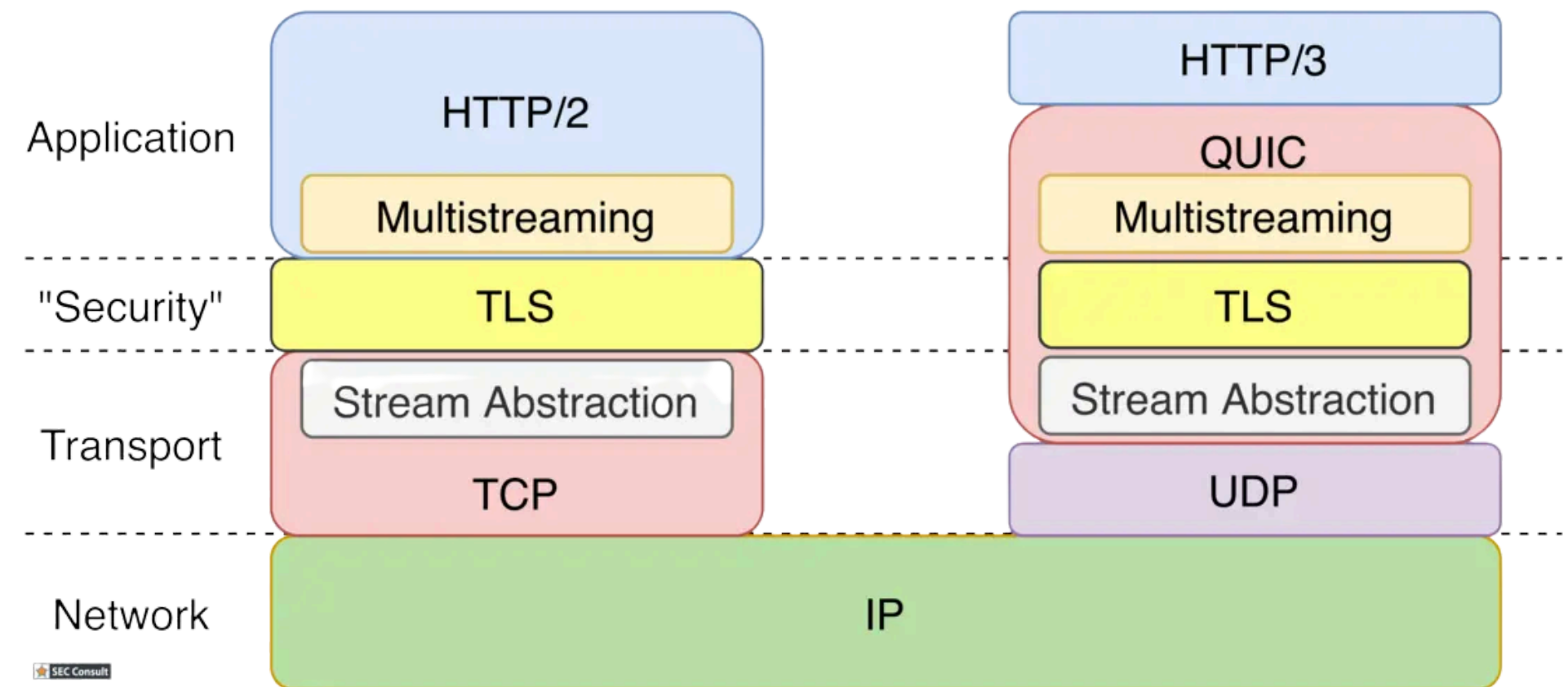
Lecture Roadmap

- Understanding HTTP over TLS: HTTPS
- Cryptography and its goals
- Encryption and its types
- TLS handshake Flow
- CA's, Chain of trust
- Discover HTTPS in AWS

Recap



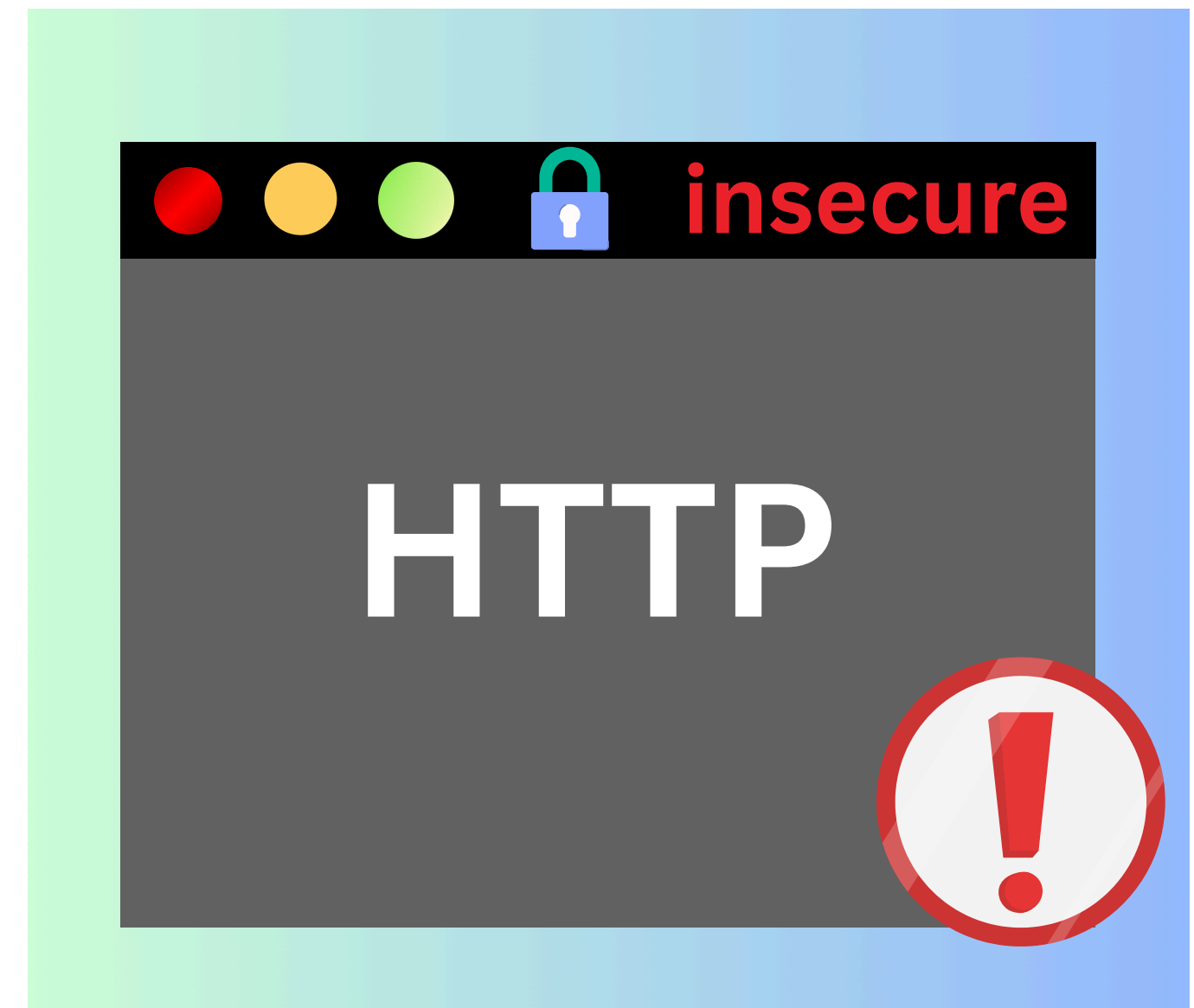
HTTP/1.0
Plain text Data



Data With TLS(Transport Layer security)

HTTP plain text by design

- HTTP lacks built-in security, making it vulnerable unless using HTTPS.
- Anyone that can capture the traffic can figure out the content of the communication between the client and the server.



HTTPS = HTTP + TLS

- HTTPS is a secure version of HTTP for safe communication over networks.
- It used SSL(Secure Socket Layer) historically
- These using TLS(Transport Layer Security) protocols to encrypt data between client and server, enhancing security compared to HTTP.



SSL/ TLS

The Padlock Question

What does  mean in browser?

Safe website? 

Encrypted connection? 

Verified identity? 

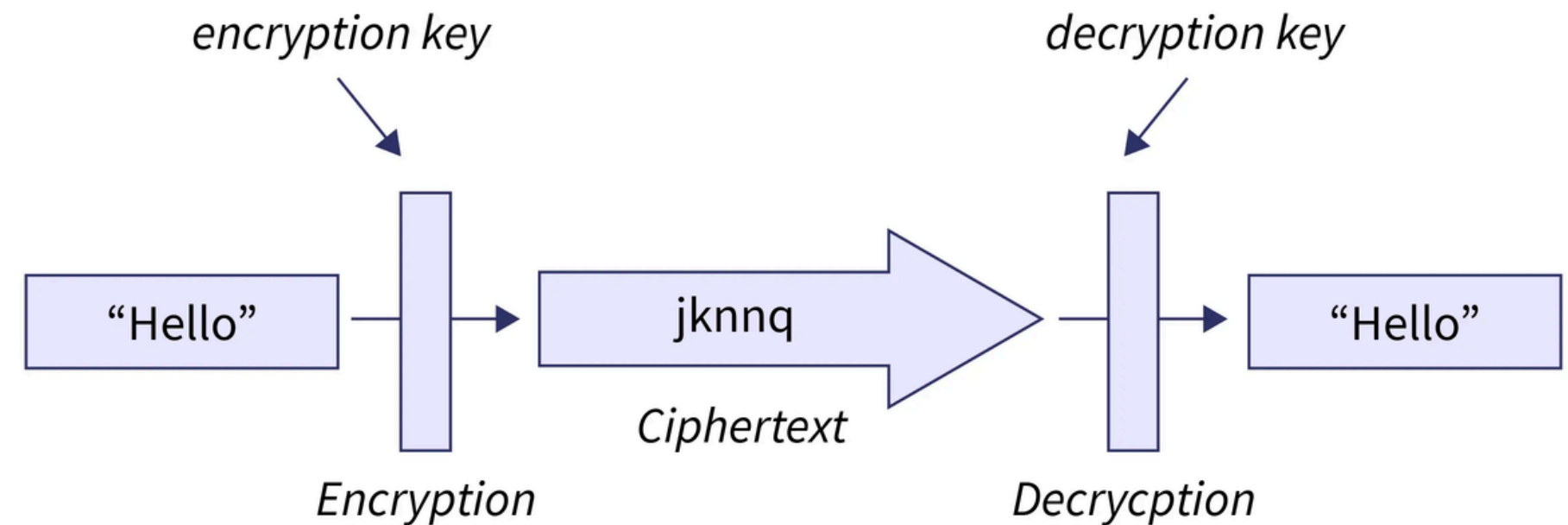
What does TLS actually provide?

Security using **Cryptography**

Cryptography

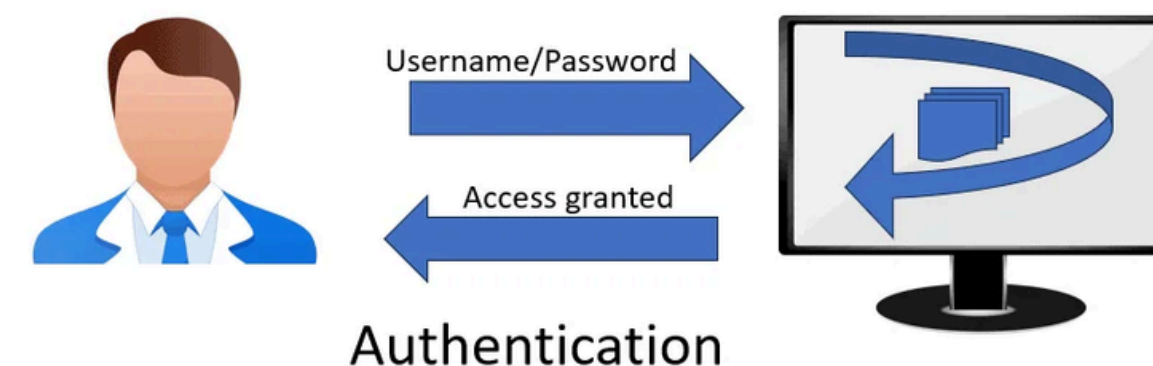
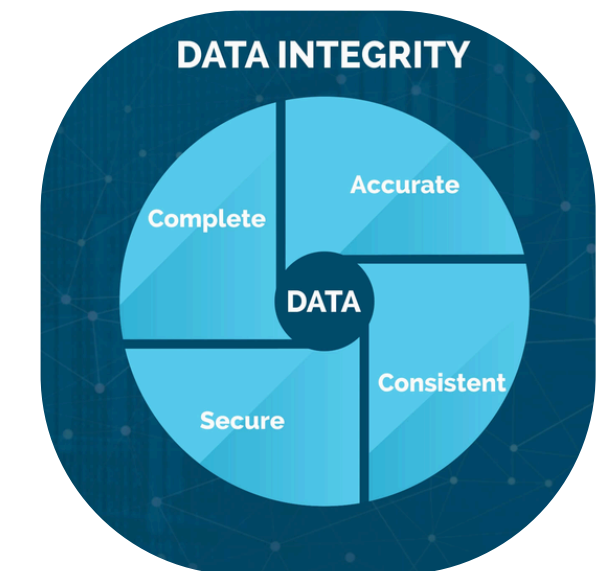
The Practice of securing information by converting readable text into unreadable code (ciphertext) and back again.

CRYPTOGRAPHY



Core Goals of Cryptography

- **Confidentiality (Privacy):** Ensures that information is accessible only to authorized parties. “**No eavesdropping**”
- **Integrity:** Guarantees that data remains accurate and hasn't been altered during transit or storage. “**No tampering**”
- **Authentication:** Verifies the identities of both the sender and the receiver. “**Verified identity**”

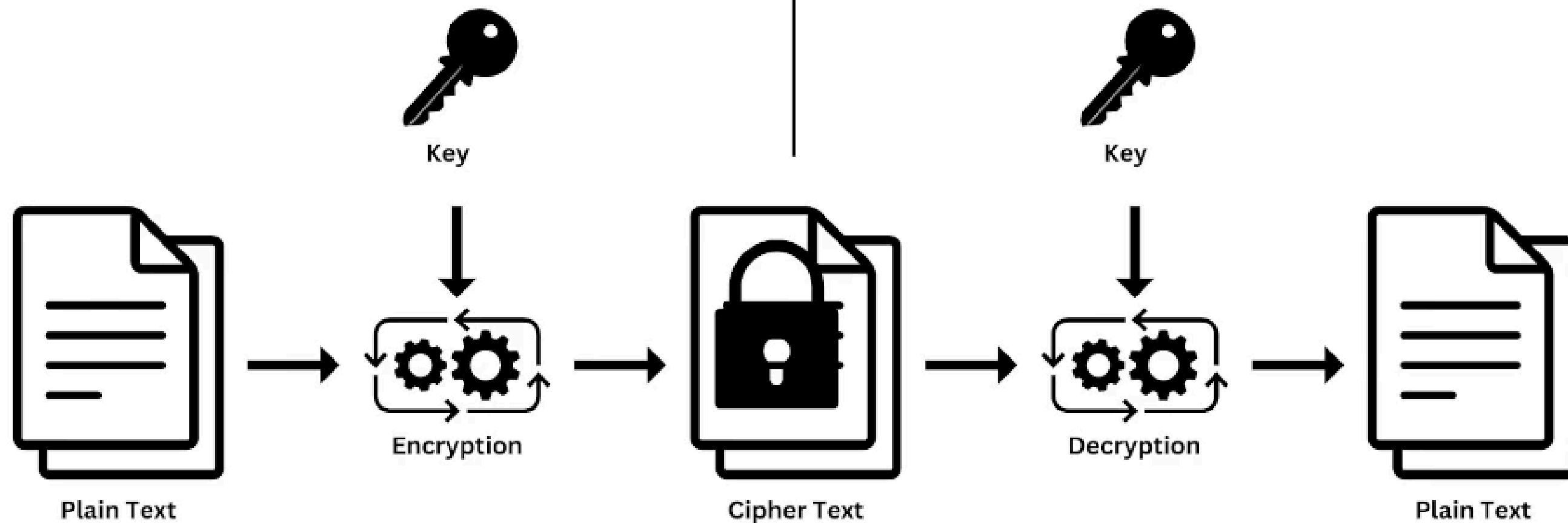


How do we achieve these goals?

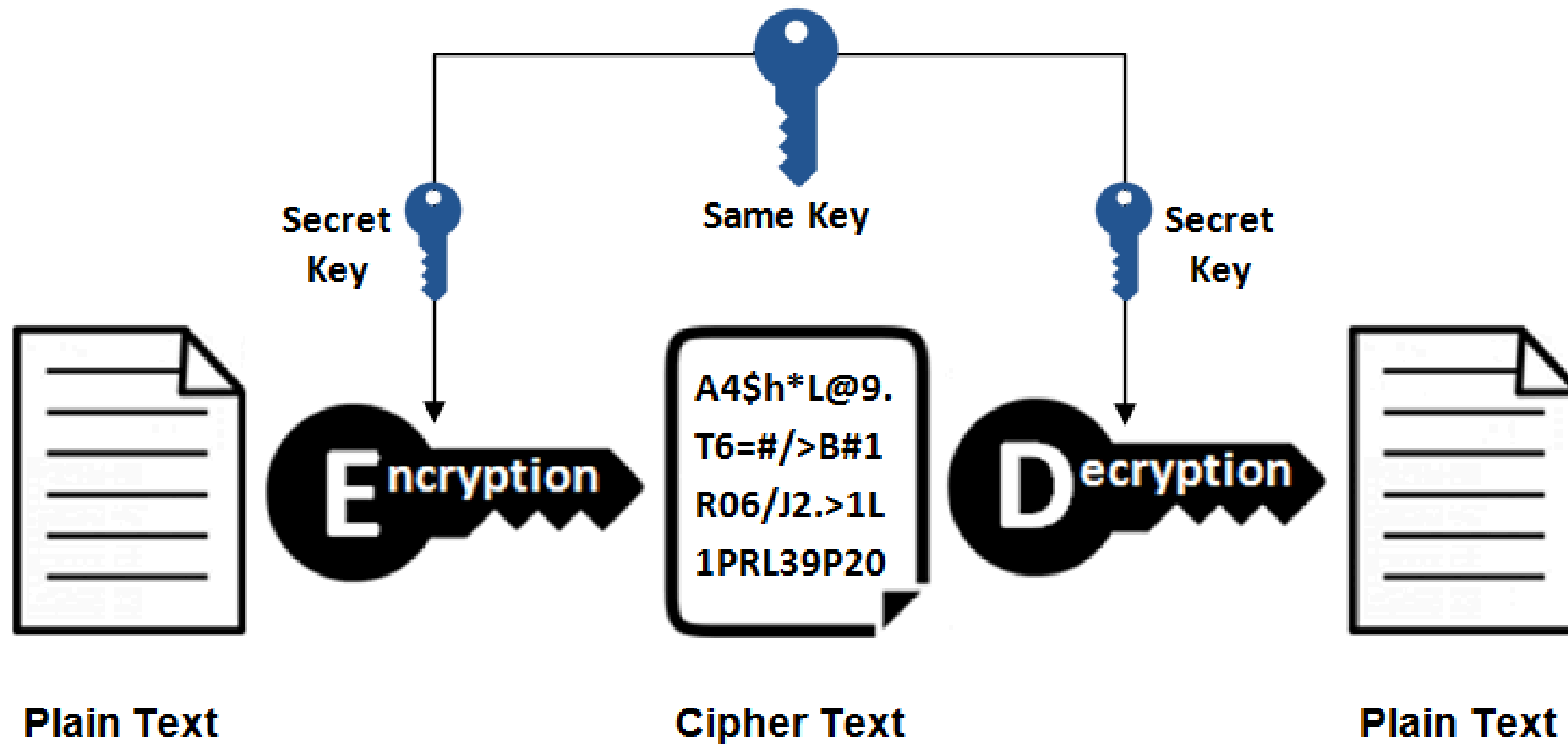
Encryption and Decryption

ENCRYPTION

DECRYPTION



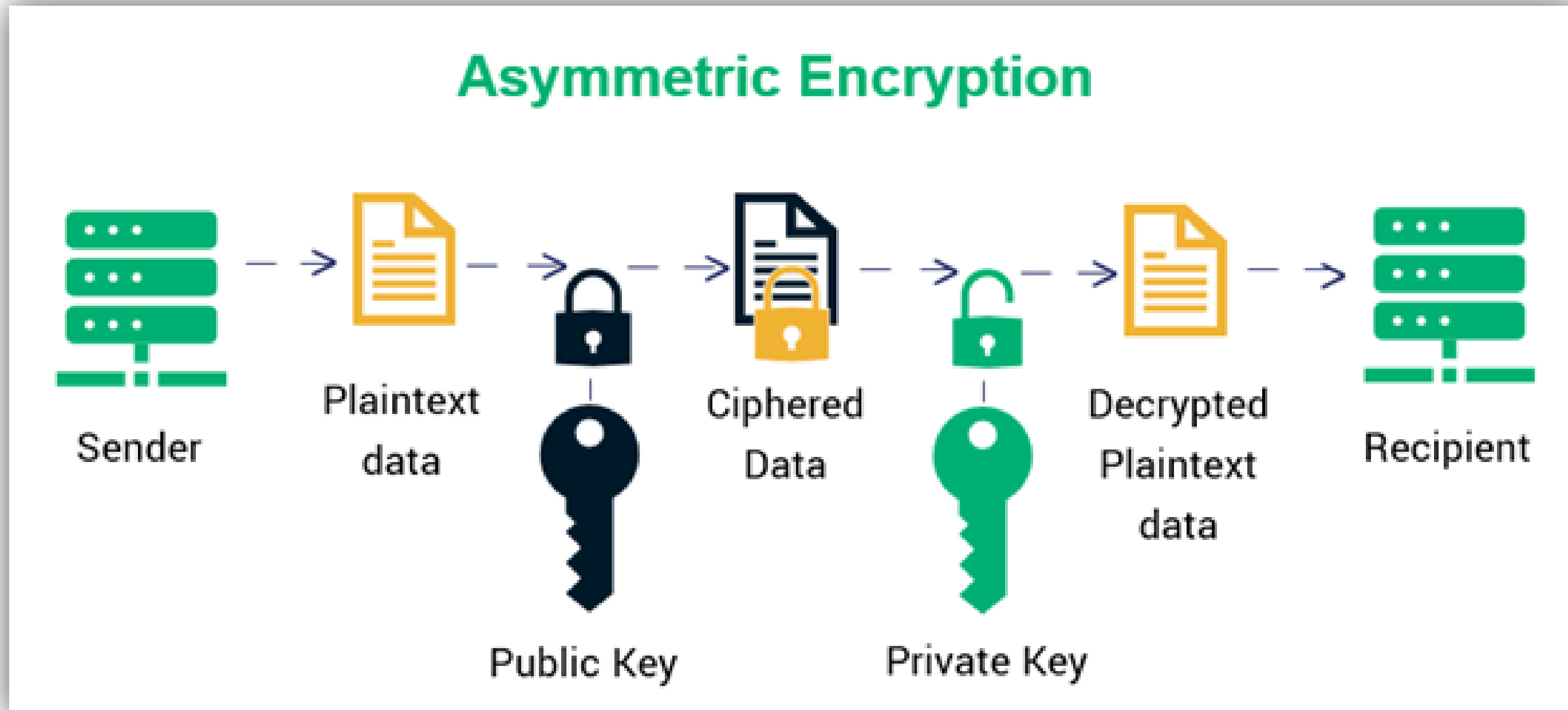
Symmetric Encryption



Analyze: whats the Issue in using symmetric encryption?

how to share the key Securely?

Asymmetric Encryption



How Do I Trust this Server?

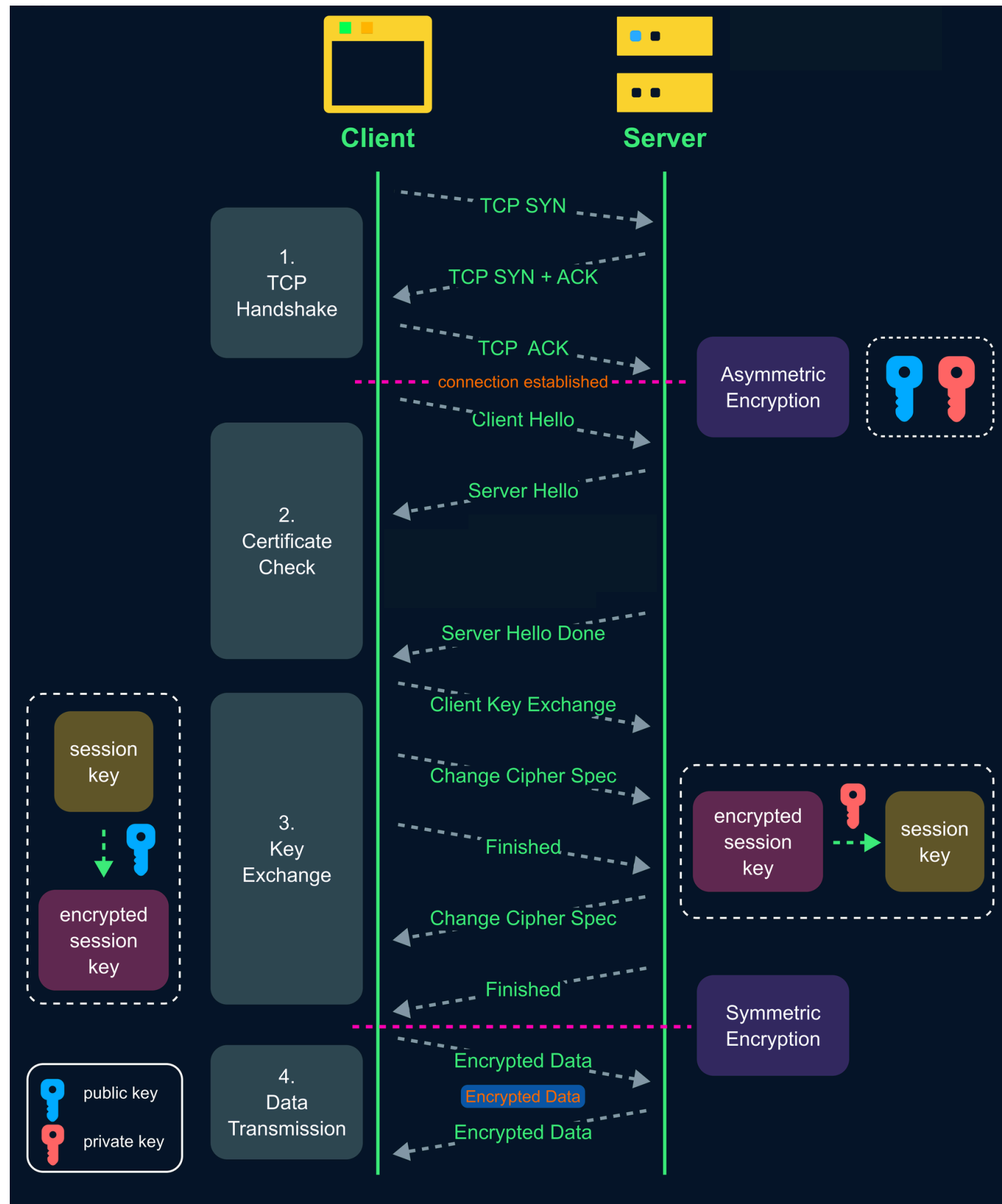
microsoft.com

VS

microsoft.com



Certificate



What is a Certificate?



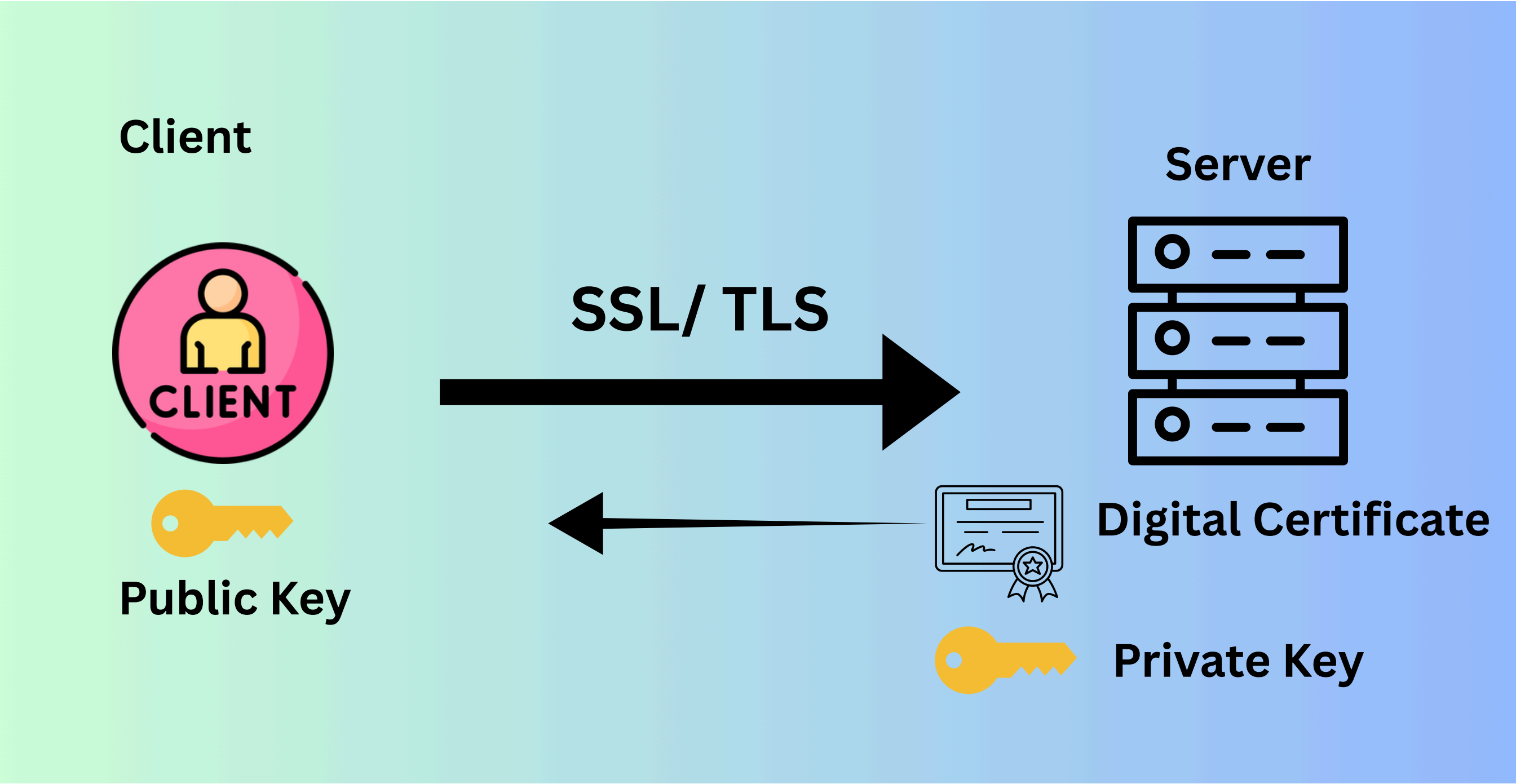
***.canva.com**
Issued by: Amazon RSA 2048 M04
Expires: Sunday, 20 December 2026 at 5:29:59 AM India Standard Time
✔ This certificate is valid

> Trust

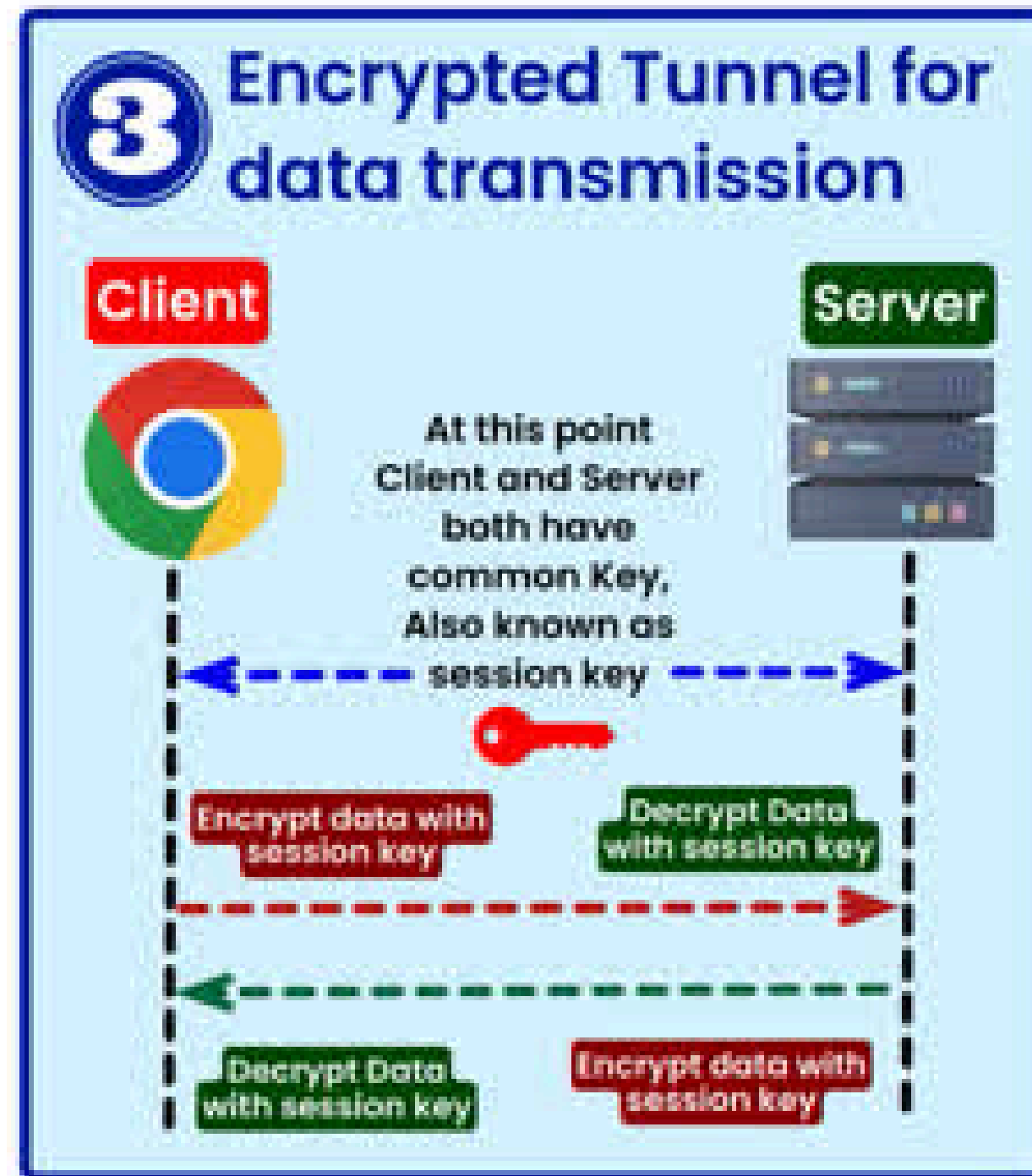
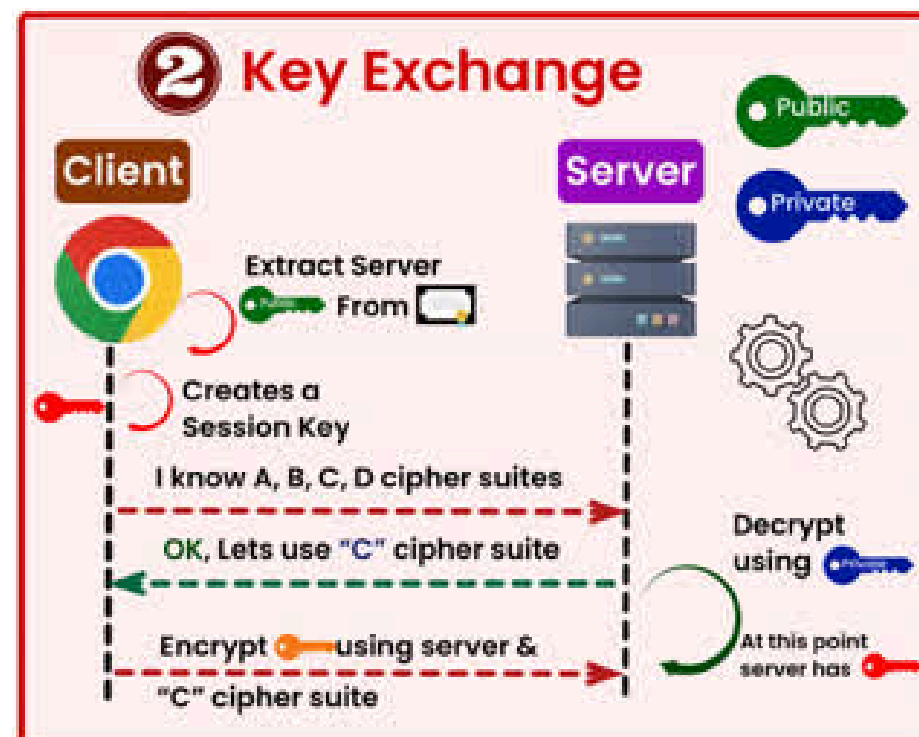
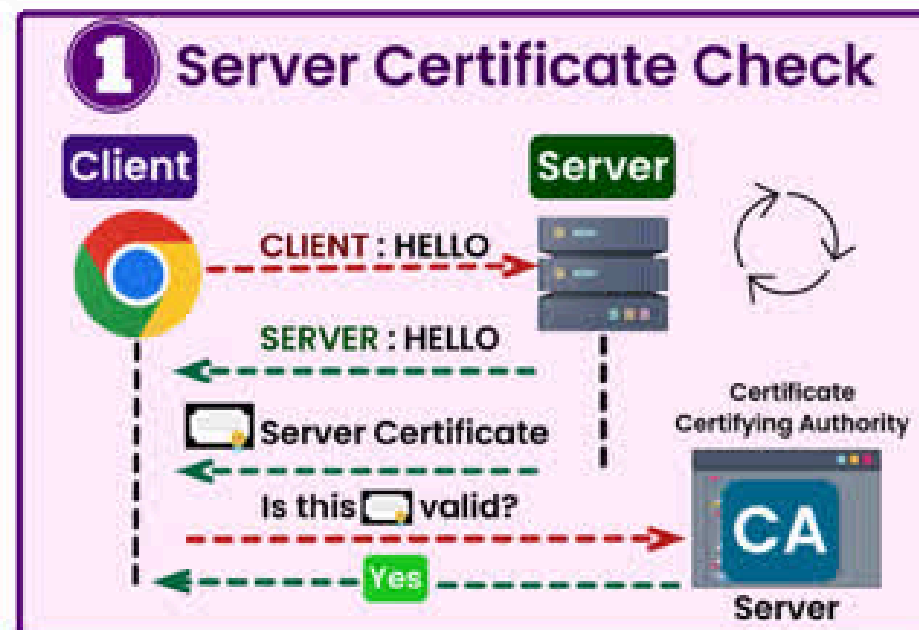
▼ Details

Subject Name	
Common Name	*.canva.com
Issuer Name	
Country or Region	US
Organisation	Amazon
Common Name	Amazon RSA 2048 M04
Serial Number	0A C6 A0 8B 42 20 3F BD 14 61 62 3D 4C 81 0B 8C

How HTTPS Works



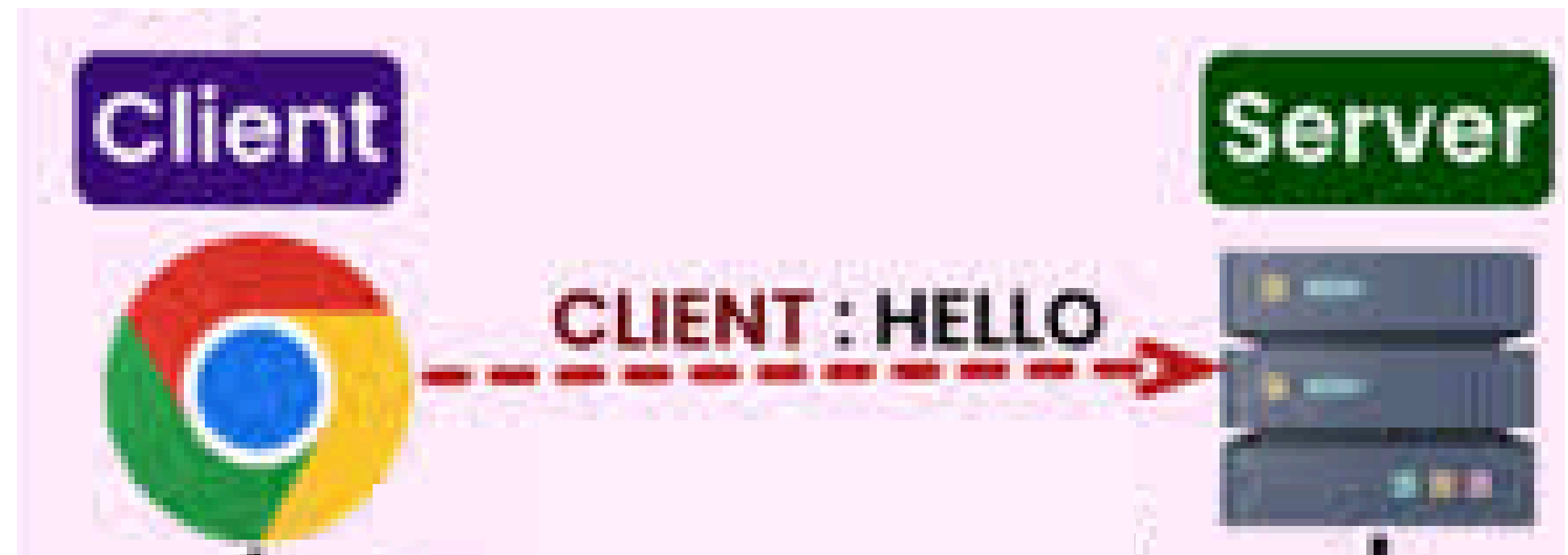
How HTTPS Works



TLS Handshake Overview

Step 1: ClientHello

Client offers TLS versions,
cipher suites, random nonce.



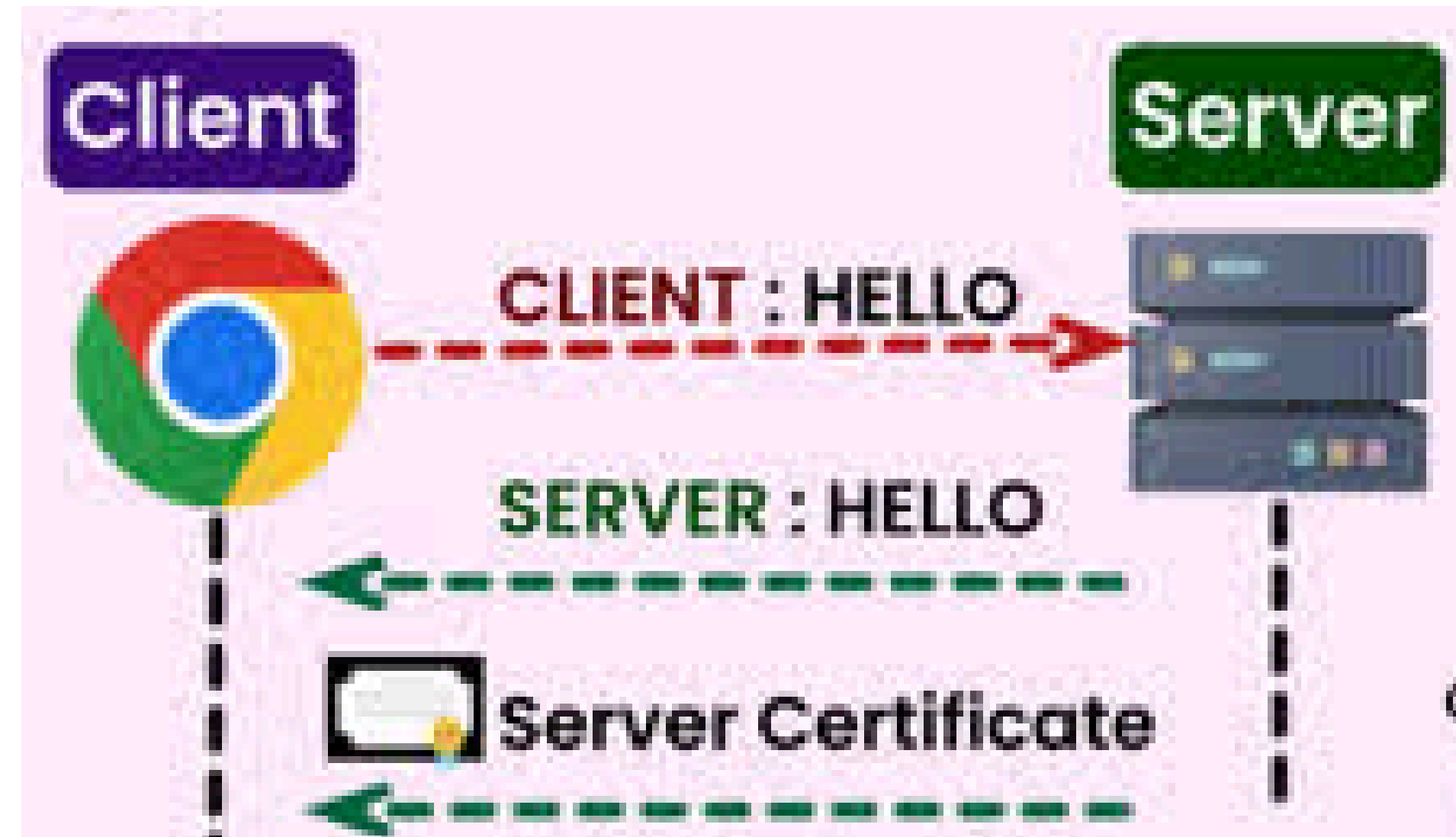
*“I can speak TLS 1.3 or TLS 1.2.
I will use XYZ algorithm for PQR purposes.
This is Number once a 32-byte string.”*

TLS Handshake Overview

Step 2: ServerHello + Certificate

Server hello is picked cipher and its certificate (public key + CA signatures).

“Out of your options XYZ for PQR purposes, we will use this following bundle of algorithms to encrypt our session and this is my identity issued by following authority.”

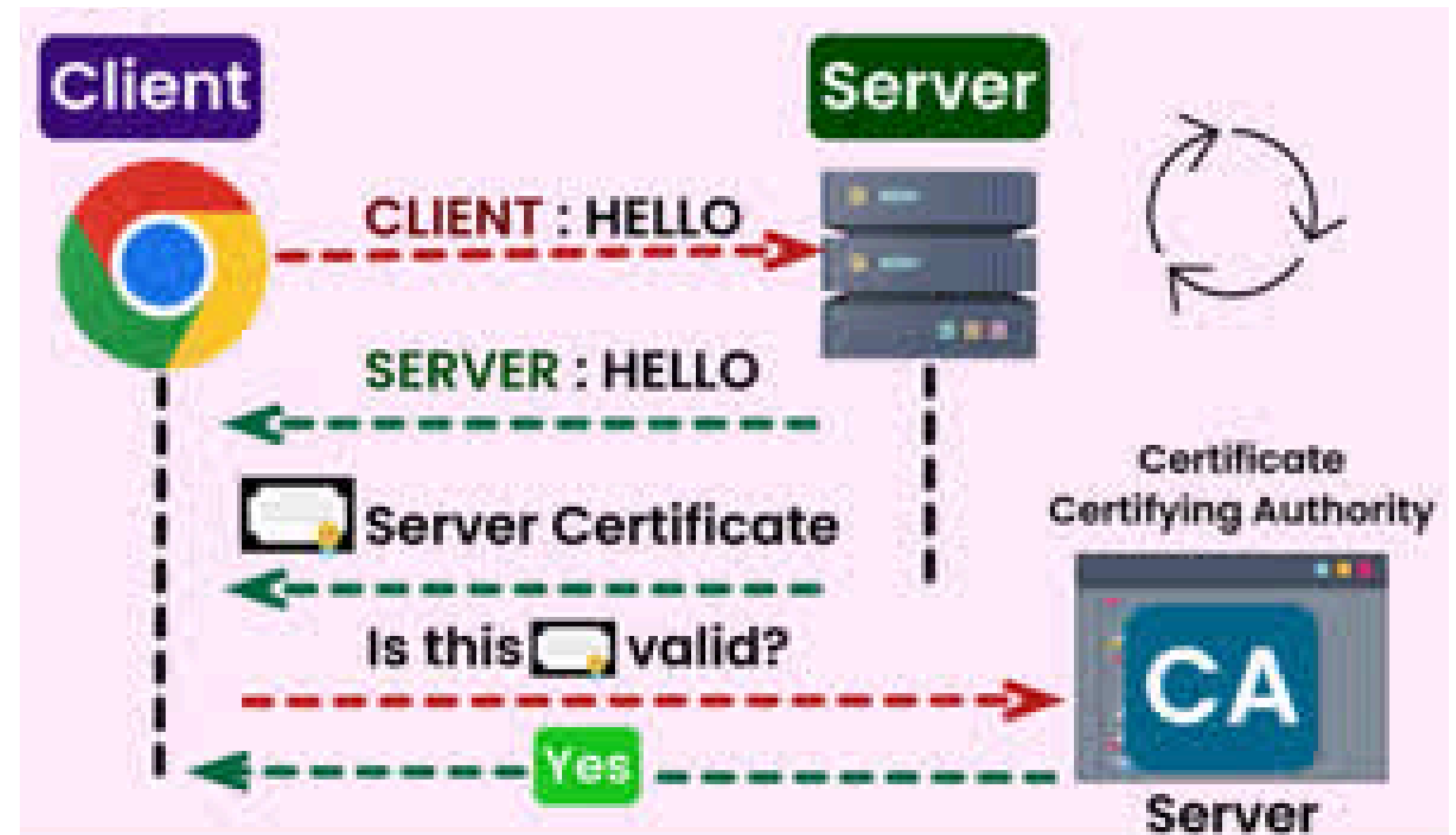


Authentication:

Step 3: ServerHello + Certificate

Client validates the certificate chain. This will stop Man in the middle attack.

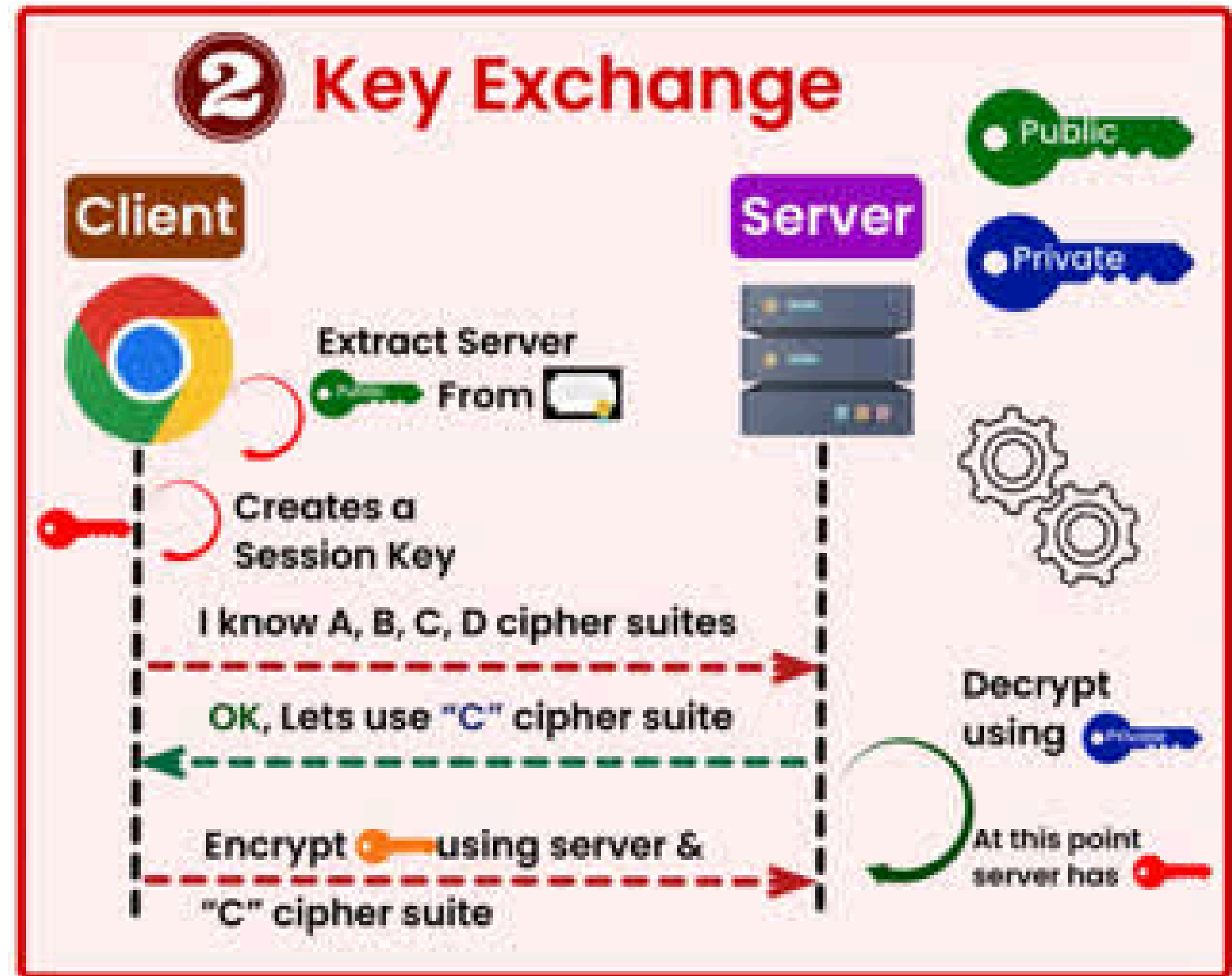
“Send request for certificate validation to CA” if its valid client can trust if not client shows website is not secure.”



Key Exchange:

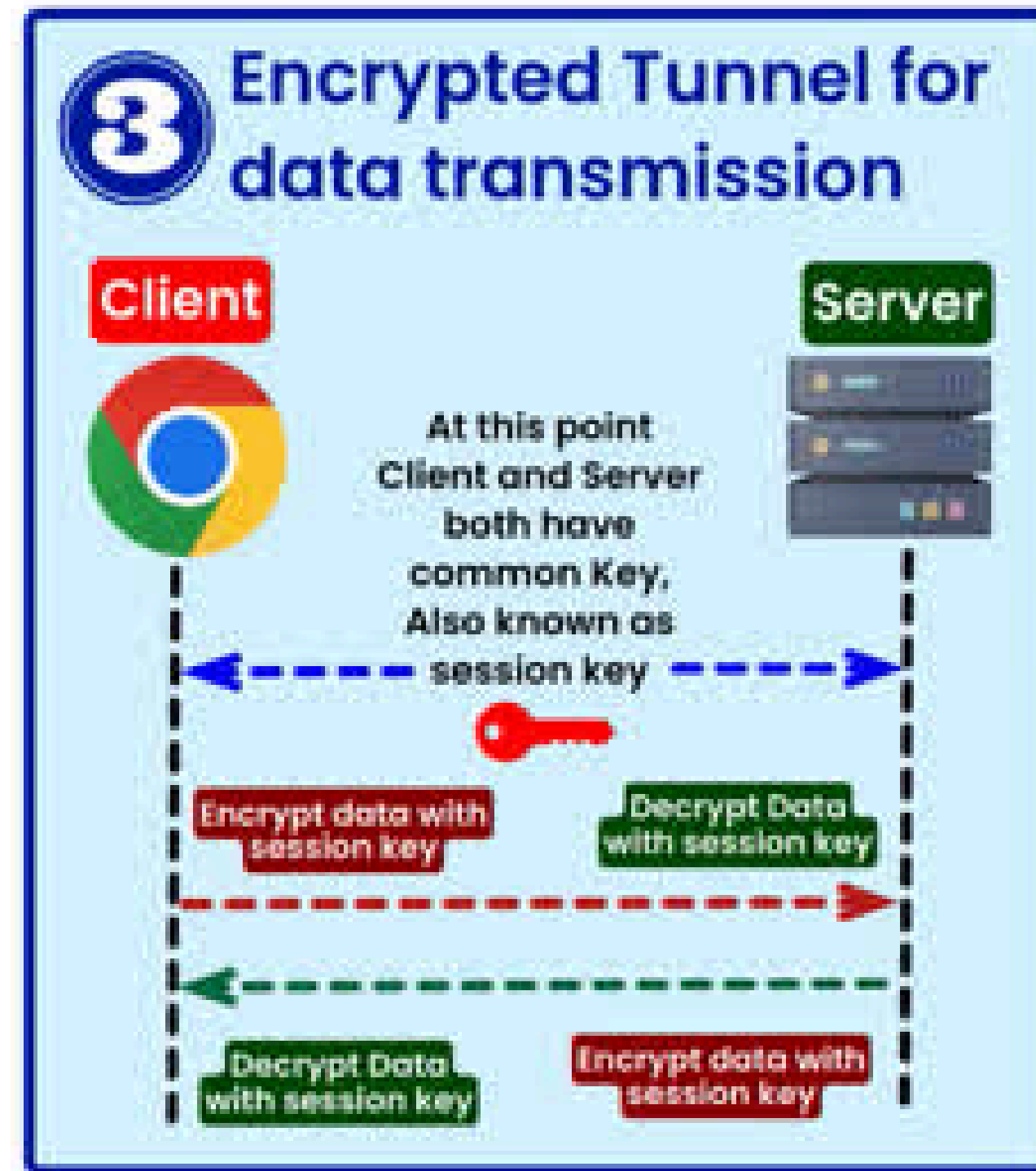
Step 4: Key Exchange

Client and Server derive a shared session key



Data Sharing on Secure Channel

Step 4: Communication Start with Encrypted application data being shared.



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